Exact Clifford Compression and Tensor Certificates on a K3 Quadratic Locus

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Abstract

We study the specified even-Clifford tensor in *Algebraic Kuga–Satake correspondences and Hodge conjectures on a K3 quadratic locus*. At a Picard jump we replace an existential Hodge retraction by a canonical coefficient projection, prove its compatibility with the prescribed right vector and both polarized tensor slots, and give a denominator bound independent of exterior degree. A concrete D_4 example separates rational coordinate denominators from integral lattice index. We also give a finite generator criterion for the exact map $a \mapsto vaw$, with a quantitative defect bound and a rational-gap equality test. These are certificates for specified maps and already supplied cycles, not proofs of cycle existence. The source's coherent half sums, curve domination, geometric comparison, representation theory, and all-period specialization remain separate imported obligations. Private original period-certificate proofs are used with their rationality, full-space, and actual-cycle hypotheses intact. Contribution assessment: C3/5, explicit functorial compression with controlled denominators and a generator-based exact-tensor certificate notably improve local verifiability while the geometric algebraicity obligations remain imported.

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1 The specified object and the scope

The assigned source [1] concerns projective complex K3 surfaces S for which the transcendental quadratic space

$$T_S = \mathrm{NS}(S)^{\perp}_{\mathbb{Q}} \subset H^2(S,\mathbb{Q})$$

with its actual cup-product form embeds isometrically over $\mathbb{Q}$ in

$$V_P = \mathbb{U}_{\mathbb{Q}}^{\oplus 2} \perp \langle -1\rangle^4, \qquad P = \mathbb{U} \oplus D_8(-1) \oplus D_4(-1). \tag{1}$$

Here $\mathbb{U}$ has Gram matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The space V_P has dimension eight and signature $(2,6)$. This is not a rank-only criterion: a rational quadratic embedding is required. In particular an integral primitive embedding, a rational isometry, a Hodge isometry, and an isomorphism of abstract vector spaces are four different assertions.

We use $\mathrm{Cl}(U,q)$ with $v^2 = q(v,v)$ and $uv + vu = 2q(u,v)$. Let $W_U = \mathrm{Cl}^+(U,q)$. For an oriented orthonormal positive period plane with basis e, f, the weight-one Hodge structure on W_U has complex structure L_J, where $J = ef$ and $J^2 = -1$. For a prescribed rational w with $q(w,w) \neq 0$, the source map is

$$j_w : U(1) \longrightarrow \mathrm{End}(W_U), \qquad j_w(v)(a) = vaw. \tag{2}$$

The Tate object $\mathbb{Q}(1)$ has weight -2. If E is a rational polarization, the identification $E^{\flat} : y \mapsto E(y,-)$ converts the second, dual slot of $\mathrm{End}(W_U) = W_U \otimes W_U^{\vee}$ to obtain $\iota_{w,E} : U \to W_U \otimes W_U$. The right vector need not be of type $(1,1)$:

$$L_g j_w(v) L_g^{-1} = j_w(gvg^{-1})$$

for left spin multiplication. Thus (2) is Hodge without that additional restriction.

The source asserts algebraicity of this *specified* map on the locus (1), and derives rational HC and GHC in every degree of every self-power. Those global statements are not established by a small residual, an injective finite matrix, a rational projector, or an unspecified algebraic summand. Our proved improvements concern exact comparison and compression at finite steps. Each conditional cycle consequence below requires its stated geometric input.

2 Canonical compression at a Picard jump

Let $U = T \perp M$ be a nondegenerate rational quadratic decomposition. Assume that the positive period plane lies in $T_{\mathbb{R}}$. At the source's ample-P Picard jumps, $M = U \cap \mathrm{NS}(S)_{\mathbb{Q}}$ is negative definite, by the Hodge index theorem, and $T = T_S$. Write $i : \mathrm{Cl}^+(T) \hookrightarrow \mathrm{Cl}^+(U)$ for the natural inclusion.

Let

$$c_U : \bigwedge U \longrightarrow \mathrm{Cl}(U), \qquad v_1 \wedge \cdots \wedge v_r \longmapsto \frac{1}{r!} \sum_{\sigma} \mathrm{sgn}(\sigma) v_{\sigma(1)} \cdots v_{\sigma(r)}$$

be the canonical PBW vector-space isomorphism. For an orthogonal basis, the images of increasing wedge monomials are simply their ordered Clifford products. PBW degree is not Clifford multiplicative degree: products can have lower-degree contractions. Let $\pi_T : U \to T$ be the q-orthogonal projection.

Theorem 2.1 (Explicit Clifford conditional expectation)**.** *The rational map*

$$\Pi_{T,U} = c_U \left(\bigoplus_{r=0}^{\dim U} \bigwedge^{r} \pi_T \right) c_U^{-1} \tag{3}$$

is an idempotent onto the included $\mathrm{Cl}(T)$*, preserves parity, and satisfies*

$$\Pi_{T,U}(xay) = x \Pi_{T,U}(a) y \quad (x, y \in \mathrm{Cl}(T),\ a \in \mathrm{Cl}(U)). \tag{4}$$

Its restriction to even parts, followed by i^{-1} *on its image, is a rational Hodge retraction* $p_{T,U} : W_U \to W_T$*. In particular*

$$p_{T,U} j_w^U(v) = j_w^T(v) p_{T,U} \qquad (v, w \in T).$$

In any orthogonal basis adapted to $T \perp M$*, it is the diagonal zero-one mask retaining precisely monomials with no* M *factor.*

Proof. Choose a rational orthogonal basis $m_1, \ldots, m_d$ of M. The reflection r_a changing the sign of m_a and fixing its orthogonal complement is a rational isometry. Its Clifford automorphism α_a changes the sign of a PBW monomial exactly when that monomial contains m_a. The α_a commute, and

$$\Pi_{T,U} = \prod_{a=1}^{d} \frac{1 + \alpha_a}{2} = 2^{-d} \sum_{S \subset \{1,\ldots,d\}} \alpha_S \tag{5}$$

is the stated mask. Equivariance of PBW under orthogonal maps shows that this is exactly (3); hence the result is independent of the chosen orthogonal basis of M.

Each α_a fixes every element of $\mathrm{Cl}(T)$, so averaging its multiplicative identity $\alpha_a(xay) = x\alpha_a(a)y$ proves (4). Every α_a fixes the period bivector $J \in \mathrm{Cl}^+(T_{\mathbb{R}})$, and consequently commutes with L_J. The even restriction is therefore a Hodge map and is the identity on W_T. For $v, w \in T$, equation (4) applies even though v, w are odd and a is even, proving the last assertion. □

The reflection-mask candidate was supplied by the project reviewer. The proof, implementation certificates, degree-independent denominator refinement, and worked lattice case here are independently developed. Finite-group averaging and PBW are classical. Neither their use nor the existence of a Hodge retraction is claimed historically new. The improvement is to make the source's abstract retraction at lines 5047–5133 explicit, auditable, and composable.

Corollary 2.2 (Nested restrictions)**.** *If $T \subset U \subset V$ are nondegenerate rational subspaces with the period plane in T, then*

$$p_{T,U}p_{U,V} = p_{T,V}.$$

Thus compression is independent of the chosen nested chain of Picard enlargements.

Proof. The rational orthogonal decomposition is $V = T \perp (U \cap T^{\perp}) \perp U^{\perp}$. The two successive masks and the one-step mask retain exactly the same monomials. □

2.1 Denominators without a degree penalty

For $m \in U$ let ϵ_m denote exterior multiplication by m and let ι_m be contraction with the covector $q(m, -)$. They satisfy $\iota_m\epsilon_m + \epsilon_m\iota_m = q(m, m)\,\mathrm{id}$.

Proposition 2.3 (Degree-independent denominator bound)**.** *In exterior coordinates the mask for an orthogonal rational basis of M is*

$$\bigoplus_r \bigwedge^r \pi_T = \prod_{a=1}^{d} \left(\mathrm{id} - \frac{\epsilon_{m_a}\iota_{m_a}}{q(m_a, m_a)}\right). \tag{6}$$

Suppose a declared ambient lattice L has integral bilinear form q, and the m_a are integral vectors in it. The matrix of this operator on $\bigwedge L$ has denominator dividing $\prod_a |q(m_a, m_a)|$, simultaneously in every degree. It incurs no additional factor 2^d from averaging and no power of this denominator proportional to exterior degree.

Proof. Decompose an exterior vector uniquely as $\omega = \omega_0 + m \wedge \omega_1$ with $\omega_0, \omega_1 \in \bigwedge m^{\perp}$. Contraction gives $\iota_m\omega = q(m, m)\omega_1$, so the factor displayed in (6) kills exactly the terms containing m. Orthogonality makes these projections commute and proves the product. In a lattice wedge basis both ϵ_m and ι_m have integral entries, proving the denominator statement. □

This is a bound in the specified exterior lattice. To use an unrelated Clifford, cohomology, or numerical period basis, the rational PBW and basis-change matrices and their inverses must also be accounted for. In an integral orthogonal basis PBW introduces no extra denominator. More generally the recursive identity $c(v \wedge \omega) = v\,c(\omega) - c(\iota_v\omega)$ shows that integral bilinear contractions preserve the integral PBW lattice in an ordered integral basis. Clearing denominators in the vectors m_a before applying the proposition is legitimate, but may produce a deliberately nonminimal bound.

2.2 The projection is not an algebra homomorphism

The bimodule identity (4) has two factors in $\mathrm{Cl}(T)$; it does not hold for arbitrary ambient factors. In a negative orthonormal coordinate plane with basis e_3, e_4, take $m = e_3 + 2e_4$, so $q(m, m) = -5$. The vector projection is

$$\pi_T|_{\langle e_3, e_4\rangle} = \frac{1}{5}\begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}.$$

Consequently $\Pi(e_3e_4) = 0$ although $\pi_T(e_3)\pi_T(e_4) = 2/5$. With an orthogonal positive unit vector $e_1 \in T$, both $a = e_1e_3$ and $b = e_1e_4$ are even and

$$\Pi(ab) = 0, \qquad \Pi(a)\Pi(b) = -2/5.$$

Thus even an exact Clifford compression must not be implemented by blind substitution into a product of projected generators.

3 Both tensor slots and prescribed polarizations

In this section all matrices act on column vectors. A polarization is represented by a nonsingular skew-symmetric matrix K with $E(x,y) = x^T K y$. Its flat map $E^\flat$ has matrix $K^T = -K$. An endomorphism with matrix A has tensor form with coefficient matrix

$$C_E(A) = AK^{-1}. \tag{7}$$

Indeed conversion of the dual factor multiplies the coefficient matrix on the right by $((K^T)^{-1})^T = K^{-1}$. This explicit convention prevents an unnoticed transpose or sign.

Proposition 3.1 (Two-slot identity and functoriality)**.** *Let $i : W_T \hookrightarrow W_U$ be a rational Hodge inclusion, $p : W_U \to W_T$ any Hodge retraction, and K_U, K_T prescribed rational polarizations. Define*

$$b = K_T^{-1} i^T K_U. \tag{8}$$

For every endomorphism A of W_U,

$$p\, C_{E_U}(A)\, b^T = C_{E_T}(pAi).$$

For nested inclusions, the corresponding second-slot maps compose exactly. Together with Corollary 2.2, canonical Clifford compression of both tensor slots is independent of intermediate polarizations and of the nested chain.

Proof. Skew-symmetry gives $b^T = K_U i K_T^{-1}$, so the asserted matrix identity is $pAK_U^{-1}K_U i K_T^{-1} = pAiK_T^{-1}$. For inclusions i_{01}, i_{12},

$$(K_0^{-1} i_{01}^T K_1)(K_1^{-1} i_{12}^T K_2) = K_0^{-1}(i_{12} i_{01})^T K_2.$$

This proves the second assertion and explicitly cancels the intermediate polarization. The first assertion is the source's compression identity; its use with the canonical mask and its composability make the local implementation definite. □

It is generally wrong to use p on both slots. Even $K_T = i^T K_U i$ is not by itself sufficient for that substitution: one must also have $p = b$. The polarization-orthogonal retraction is

$$p_E = (i^T K_U i)^{-1} i^T K_U. \tag{9}$$

The restricted form is nondegenerate because it polarizes the sub-Hodge structure, and $p_E i = \mathrm{id}$. Orthogonal complement with respect to the associated positive Hermitian form is Hodge, so p_E is a rational Hodge map. It need not equal the Clifford mask. If $K_T = i^T K_U i$ and $p = p_E$, then and only then the second-slot map in (8) equals this chosen p.

For example take $W_U = \mathbb{Q}^2 \oplus \mathbb{Q}^2$ with the same complex structure on both summands, $i(x) = (x, 0)$, and $p(x,y) = x$. Let $K_0 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and

$$K_U = \begin{pmatrix} K_0 & cK_0 \\ cK_0 & K_0 \end{pmatrix}, \qquad K_T = K_0, \qquad 0 < |c| < 1, \quad c \in \mathbb{Q}.$$

These are polarizations after one consistent sign convention for the complex structure: the associated multiplicity form $\left(\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right)$ is positive. Here $b = (I, cI) \neq p$. Already for $A = \mathrm{id}$, $pK_U^{-1}p^T = (1-c^2)^{-1}K_0^{-1} \neq K_0^{-1}$. The correct second slot restores the exact identity.

Right-vector changes and cycle consequences. For nonisotropic w_0, w put $r = w_0^{-1}w \in \mathrm{Cl}^+(U)^\times$. Then

$$j_w = R_r j_{w_0}, \qquad R_r(a) = ar. \tag{10}$$

Right multiplication commutes with L_J, so R_r is a rational Hodge automorphism. If $w \in T$, then j_w preserves W_T, whether or not R_r separately preserves it. Thus the exact formula is

$$(p_{T,U} R_r \otimes K_T^{-1} i^T K_U)\iota^U_{w_0,E_U} = \iota^T_{w,E_T}. \tag{11}$$

The order in the first slot cannot be reversed without additional invariance. A rational Hodge map between abelian H^1 spaces is induced by a rational algebraic correspondence: on dual homology clear denominators to obtain a map of integral lattices commuting with complex structures; it gives a holomorphic homomorphism of abelian varieties, algebraic by projectivity and GAGA. Consequently (11) transports an *already supplied algebraic ambient tensor* to the desired one. It does not produce that ambient cycle.

4 A generator certificate for the exact tensor

Let U be any nondegenerate rational quadratic space of dimension $n \geq 2$, with an orthogonal basis $e_1, \ldots, e_n$ and $a_i = q(e_i, e_i) \neq 0$. Put $W = \mathrm{Cl}^+(U)$ and

$$z_i = e_1 e_i \quad (2 \leq i \leq n), \qquad \sigma_w(b) = w^{-1}bw \quad (b \in W).$$

The inverse of w is $w/q(w,w)$; σ_w is a unital automorphism of the even algebra.

Theorem 4.1 (Twisted right-module test)**.** *For a rational linear operator $A : W \to W$ and a given $v \in U$, the following are equivalent:*

1. *$A(a) = vaw$ for every $a \in W$;*
2. *$A(1) = vw$ and, for every $2 \leq i \leq n$,*

$$AR_{z_i} = R_{\sigma_w(z_i)}A. \tag{12}$$

For a rational linear map $U \to \mathrm{End}(W)$ it suffices to check these identities at a rational basis of U. Thus a full-rank embedding can be distinguished from the prescribed tensor without searching for an unspecified change of target coordinates.

Proof. The standard map satisfies $j_w(v)(ab) = j_w(v)(a)\sigma_w(b)$ and its unit value is vw. Conversely the z_i generate W as a unital algebra: they include $e_1 e_i$, while for $i, j > 1$, $i \neq j$, $e_i e_j = -z_i z_j / a_1$. All even monomials are products of such pairs. Equation (12), by induction on words and then rational linearity, gives $A(ab) = A(a)\sigma_w(b)$ for every $a, b \in W$. Taking $a = 1$ gives $A(b) = vw\, w^{-1}bw = vbw$. Checking a source basis is sufficient by linearity in v. □

The unit condition cannot be dropped: all maps $a \mapsto c\,\sigma_w(a)$, for arbitrary $c \in W$, satisfy the intertwining conditions. Nor can the intertwining conditions be dropped: a nonzero linear operator vanishing at 1 can be added without changing the unit value. Even $2j_w$ has the same image and rank as j_w but is not the specified map. If a candidate is supplied in tensor coordinates $C(v)$, first use $A(v) = C(v)K$ with the stated polarization convention.

4.1 A quantitative defect bound on the actual ambient form

The rational form V_P admits an orthogonal basis with $a_i \in \{1, -1\}$, exactly two positive entries. Indeed in each hyperbolic plane with basis u, v and $q(u, v) = 1$ use $u + v/2, u - v/2$. We may order it with $a_1 = 1$. This is a rational coordinate change, not an integral lattice identification or a prescription of the period plane. Equip W with its ordered even PBW coordinate basis. For vectors use the maximum coordinate norm, and for matrices $\|A\|_{\max} = \max_{r,s} |A_{rs}|$.

Proposition 4.2 (Short-word defect and equality gap)**.** *Assume $a_i \in \{1, -1\}$ and $w = e_1$. For a candidate operator A and a specified v set*

$$\epsilon = \|A(1) - ve_1\|_\infty , \qquad \delta = \max_{2 \le i \le n} \left\| AR_{z_i} - R_{\sigma_{e_1}(z_i)} A \right\|_{\max} .$$

Then

$$\|A - j_{e_1}(v)\|_{\max} \le \epsilon + (n-1)\delta. \tag{13}$$

If all entries of this difference are independently known to belong to $H^{-1}\mathbb{Z}$, $H \in \mathbb{Z}_{>0}$, and rigorous enclosures give $\epsilon + (n-1)\delta < 1/H$, then $A = j_{e_1}(v)$ exactly.

Proof. Each even PBW monomial is, up to sign, a word of length at most $n - 1$ in the z_i. If it does not contain e_1, pair its indices and use $e_i e_j = -z_i z_j / a_1$; if it contains e_1, use one z_i first and pair the remaining indices. Every prefix of the resulting word is a signed PBW monomial. Also $\sigma_{e_1}(z_i) = -z_i$, so its right action is a signed permutation of PBW coordinates.

Let $D = A - j_{e_1}(v)$ and $\Delta_i = AR_{z_i} - R_{\sigma_{e_1}(z_i)} A$. Since the standard map has zero defect,

$$D(bz_i) = R_{\sigma_{e_1}(z_i)} D(b) + \Delta_i(b).$$

For each signed PBW prefix b, the last term has norm at most δ, and the right action preserves the maximum norm. Induction from $b = 1$ gives error at most $\epsilon + \ell\delta$ on a word of length ℓ. This proves the matrix entry bound. A nonzero entry in $H^{-1}\mathbb{Z}$ has absolute value at least $1/H$, which proves the strict-gap statement. □

For the eight-dimensional ambient form, the constant in (13) is seven and $\dim W = 128$. The seven right-generator matrices are signed permutations. For one candidate operator the intertwiner checks therefore require $O(7 \cdot 128^2)$ rational scalar operations, rather than checks against all 128 basis elements of the algebra. This is an operation count for checking a supplied matrix, not a measured running-time claim or a reduction in the effort of constructing a cycle.

An arbitrary prescribed nonisotropic w is handled exactly by normalizing the candidate to $A' = R_{w^{-1}e_1} A$. The target then becomes $j_{e_1}(v)$. To return an approximate bound to the original coordinates, multiply the right side of (13) by $\left\| R_{e_1^{-1}w} \right\|_{\infty \to \infty}$. Alternatively use the strict gap with a proved denominator for $A' - j_{e_1}(v)$ and transport the resulting exact equality back. The rational basis change, $q(w, w)^{-1}$, and polarization inverse all contribute to that denominator. No such gap is inferred from rounding decimal entries.

4.2 The actual-cycle gate

The private original proof [2], lines 587–616, gives a full-space rational lattice-gap criterion. Proposition 4.2 specializes its philosophy to a structural test that is injective on all candidate endomorphisms after adjoining the unit value, with an explicit inverse estimate. The source membership gate remains essential: a cycle on $S \times A^2$ must first be projected by known algebraic π_{T_S} and $\pi_1^A \boxtimes \pi_1^A$ if its other

Künneth components are not already known to vanish. This creates a genuine rational cycle with image in the comparison space. A numerical projection does not do so. Once an actual projected cycle and exact rational coordinates or independent denominator/enclosure data are available, the generator test proves equality to the desired tensor. It neither supplies the cycle nor determines the full rational Hodge space.

5 Worked rational and integral lattice certificates

5.1 A nonadapted D_4 Picard compression

Realize the negative lattice

$$L = D_4(-1) = \{x \in \mathbb{Z}^4 : \textstyle\sum_i x_i \equiv 0 \pmod 2\}, \qquad q(x,y) = -x \cdot y.$$

Its basis matrix, with columns in the ambient coordinates, is

$$B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \qquad |\det B| = 2, \quad \det(-B^t B) = 4.$$

Take $m = (1,2,0,0)$, $M = \mathbb{Q}m$ and $T = m^\perp$. The primitive generator of $L \cap M$ is $u = 2m$, not m: the parity condition on km is $3k \equiv 0 \pmod 2$. A basis of $L \cap T$ is

$$t_1 = (-4,2,0,0), \quad t_2 = (-2,1,1,0), \quad t_3 = (-2,1,0,1).$$

Indeed $x + 2y = 0$ and $y \equiv z + t \pmod 2$ imply uniquely $(-2y, y, z, t) = at_1 + zt_2 + tt_3$, with $a = (y - z - t)/2 \in \mathbb{Z}$. Thus the positive Gram matrix on this negative definite complement is

$$-(q(t_i,t_j)) = \begin{pmatrix} 20 & 10 & 10 \\ 10 & 6 & 5 \\ 10 & 5 & 6 \end{pmatrix}, \qquad \det = 20.$$

Writing $A = (u, t_1, t_2, t_3)$, direct substitution gives

$$B^{-1}A = \begin{pmatrix} 2 & -4 & -2 & -2 \\ 6 & -2 & -1 & -1 \\ 3 & -1 & 0 & -1 \\ 3 & -1 & 0 & 0 \end{pmatrix}, \qquad |\det(B^{-1}A)| = 10.$$

Consequently

$$[L : (L \cap M) \oplus (L \cap T)] = 10, \qquad 20 \cdot 20 = 10^2 \cdot 4. \tag{14}$$

The index square is indispensable in this discriminant calculation.

The rational orthogonal projection in ambient coordinates is

$$\pi_T = \begin{pmatrix} 4/5 & -2/5 & 0 & 0 \\ -2/5 & 1/5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In the integral D_4 basis, however, it is

$$B^{-1}\pi_T B = \begin{pmatrix} 6/5 & -2/5 & 0 & 0 \\ 3/5 & -1/5 & 0 & 0 \\ 3/10 & -3/5 & 1 & 0 \\ 3/10 & -3/5 & 0 & 1 \end{pmatrix}. \tag{15}$$

Its least common denominator is ten. The denominator five in ambient coordinates is therefore not an integral-lattice answer. Using the integral vector u of norm -20 in Proposition 2.3 gives the valid, not necessarily minimal, common bound twenty in every exterior degree. In fact the exterior matrices of (15) are idempotent and have denominators dividing twenty; these claims are also checked independently by exact minors.

This fixture sits in the negative D_4 part of the source's $T_P = \mathbb{U} \oplus \mathbb{U}(2) \oplus D_4(-1)$. It demonstrates a permissible rational negative direction and the cost of changing integral frames. It does not construct a K3 surface with a prescribed divisor or an algebraic cycle realizing that divisor. Those existence requirements remain geometric.

5.2 What the discriminant-form check does establish

For completeness, the small finite forms underlying the source's lattice normalization can be checked without period approximations. For $D_n(-1)$, $n = 4, 8$, representatives of the four discriminant classes are $0, v, s, v+s$, where $v = e_1$ and $s = (1/2, \ldots, 1/2)$. Their quadratic values modulo $2\mathbb{Z}$ are

	0	v	s	$v+s$
$D_4(-1)$	0	1	1	1
$D_8(-1)$	0	1	0	0

.

They follow by taking the negative squared norms; the displayed representatives also permit a direct check of every associated bilinear pairing modulo $\mathbb{Z}$. The $\mathbb{U}(2)$ discriminant form has values $0, 0, 0, 1$ on $0, u/2, v/2, (u+v)/2$. Thus A_P and A_{T_P} are both the orthogonal sum of a split two-dimensional form and an anisotropic two-dimensional form over $\mathbb{F}_2$, with sixteen elements. All quadratic values are integral modulo $2\mathbb{Z}$, so negation preserves them. An explicitly matched pair of these forms is consequently an anti-isometry.

The graph of that anti-isometry defines the usual integral overlattice of $P \oplus T_P$: in the dual lattice take the inverse image of the graph. Its quadratic form is even because the two discriminant values cancel; its bilinear form is integral by polarization. The graph has order sixteen, so the new determinant is $16 \cdot 16/16^2 = 1$. Each summand is primitive: a graph element whose other coordinate is zero is zero. Its signature is $(3, 19)$. Identifying this even unimodular lattice with the K3 lattice invokes the standard lattice classification used in the source. Period realization, ample chambers and the resulting geometric moduli statement are further inputs, not consequences of the sixteen-element enumeration alone.

Over $\mathbb{Q}$, rescaling one vector of $\mathbb{U}(2)$ by $1/2$ identifies it with $\mathbb{U}$, and $D_4(-1)_{\mathbb{Q}}$ is the negative unit form on $\mathbb{Q}^4$. This gives exactly V_P in (1). The rescaling is not unimodular over $\mathbb{Z}$. The distinction between a rational isometry and an integral primitive embedding is retained throughout.

6 Error, height and actual-cycle ledgers

The exact formulas permit numerical assistance, but only with declared arithmetic data. We record the limits explicitly.

6.1 Conditioning of the three finite operations

In an adapted PBW coordinate basis the mask Π is diagonal with entries zero and one, hence has norm at most one for both the Euclidean norm and the maximum norm. These are coordinate norms, not the indefinite quadratic form on U. Under a change of Clifford frame S, its norm is bounded by $\|S\| \left\|S^{-1}\right\|$. Analytic preconditioning can reduce this condition number while increasing rational heights and denominators; these are distinct objectives.

For $b = K_T^{-1} i^t K_U$ the relevant coercivity parameter is a lower bound for the smallest singular value of the skew polarization matrix K_T, not a purported positive eigenvalue. If $K_T \widehat{b} - i^t K_U = R$ is an exactly enclosed residual and $\sigma_{\min}(K_T) \geq s > 0$, then

$$\left\|\widehat{b} - b\right\|_2 \leq \|R\|_2 / s.$$

With uncertain matrix data one must also pay their errors. For example, if X solves $BX = P$, and $\left\|B - \widehat{B}\right\| \leq e_B$, $\left\|P - \widehat{P}\right\| \leq e_P$, $\left\|\widehat{B}\widehat{X} - \widehat{P}\right\| \leq e_{\text{solve}}$, and $\sigma_{\min}(\widehat{B}) \geq s_B > e_B$, then

$$\left\|X - \widehat{X}\right\| \leq \frac{e_P + e_B \left\|\widehat{X}\right\| + e_{\text{solve}}}{s_B - e_B}. \tag{16}$$

Indeed $B(X - \widehat{X})$ is the sum of the three indicated errors, and $\sigma_{\min}(B) \geq s_B - e_B$. This is the finite solve estimate in the inspected private original proof, specialized here without an unproved infinite-dimensional inverse bound.

For any compatible finite matrices, exact and approximate compressed tensor matrices obey

$$\begin{aligned}\left\|pCb^t - \widehat{p}\widehat{C}\widehat{b}^t\right\|_2 \leq{}& \delta_p \|C\|_2 \|b\|_2 + \|\widehat{p}\|_2 \delta_C \|b\|_2 \\ &+ \|\widehat{p}\|_2 \left\|\widehat{C}\right\|_2 \delta_b,\end{aligned} \tag{17}$$

where each δ bounds the corresponding spectral-norm error. Insert and subtract $\widehat{p}Cb^t$ and $\widehat{p}\widehat{C}b^t$ to prove the inequality. If a right-vector change is used, its operator and enclosure errors enter before this compression. No cancellation of those costs is assumed.

6.2 Arithmetic gates

In rational coordinates, denominators of a product divide the product of known denominators of its factors. Matrix inverses require the determinant and adjugate, after clearing denominators. Accordingly a certificate must record the chosen quadratic and Clifford frames, $q(w_0, w_0)^{-1}$ for w_0^{-1}, both polarization matrices, any isogeny lattice multiplier and the ambient cycle's rational coefficient denominators. The denominator bound for Π does not erase any of these independent quantities.

The equality gap in Proposition 4.2 is strict. For any integer $H > 0$, a nonzero matrix unit divided by H has maximum norm exactly $1/H$, so a weak inequality would be false. Nor does a small unmeasured remainder become zero merely because all displayed coordinates are small. The structural generator test sees the whole endomorphism space once the unit value is included. A truncated period frame needs a separate injectivity or tail theorem before it can replace that test.

Finally, rational cycle classes on a *fixed* variety form a rational linear subspace of finite-dimensional cohomology. Its real span is closed, and a rational class in that real span lies in the rational subspace, by extension of a rational basis. Thus convergence of actual cycle classes in the whole fixed cohomology space can sometimes prove membership. Convergence of only selected moments, convergence on changing varieties, or approximation by formal Hodge tensors does not satisfy that argument. This is the actual quotient-space gate in the inspected private proof; it is not an analytic substitute for constructing the source's correspondences.

7 Source architecture and retained obligations

The assigned build consists of a single 5,391-line manuscript. Its entire section and theorem architecture was mapped. Literal inspection covered 3,465 distinct lines, including the bibliography, in the intervals 1–1484, 1623–1680, 1794–2410, 3277–3334, and 4144–5391. The complementary intervals are 1485–1622, 1681–1793, 2411–3276, and 3335–4143. The following account distinguishes inspected deductions from contracts whose internal geometric proofs were not inspected. Reading a cited statement or a bibliography is not reading all proofs on which it depends.

7.1 From one exact tensor to all powers

The source's first major implication, lines 366–1207, starts with the actual cycle for $\iota_{w,E}$, not merely a Hodge subspace isomorphic to T_S. The Neron–Severi intersection matrix gives an algebraic transcendental projector, and the ordinary alternator on the even-degree K3 factors gives $\bigwedge^k T_S$. Antisymmetrized compositions of j_w define an injective map: evaluation at 1 is Clifford antisymmetrization followed by the invertible factor w^k. Thus PBW and nonisotropy of w prove injectivity for the prescribed map.

The resulting correspondence from S^k to $Z = A^2$ has codimension $k+1$. Its image is a primitive degree-two abelian component, with the required Tate shift. The polarization supplies a transpose return; nondegeneracy on the image makes the composite invertible, and a polynomial in that composite supplies its inverse. This uses Cayley–Hamilton on a finite space, not an assumed algebraic inverse Lefschetz operator in all degrees. The return correspondence has codimension $\dim Z + k - 1$, giving support in codimension $k-1$; type (k,k) classes gain the divisor step needed for codimension k.

The passage to all tensor powers additionally imports the classification of the K3 endomorphism field and Hodge group. In the totally real case, relative volume generators cannot be discarded in favor of contractions alone. In the CM case, norm-one field elements span the required endomorphisms, with rational Hodge isometries realized algebraically. Orthogonal and general-linear invariant theory then organize the tensors. For GHC, rational highest-weight carriers and their support bounds must be retained; the width across embeddings is a maximum, not a sum of independent widths. This is a substantive representation-theoretic and geometric chain. Our finite Clifford tests do not certify those imported global theorems.

7.2 The double cover, coherent half sums and domination

The inspected double-cover construction, lines 1208–1484, starts from a principally polarized abelian surface and an order-four character. Two involutions isolate an eigenspace with signature $(2,6)$. The resolved quotient has a $(2,14)$ lattice with eight orthogonal algebraic vectors of square -4. Their complement carries the relevant form q_Y. The pullback to the covering surface scales the form:

$$q_X(\beta x, \beta y) = 2q_Y(x,y).$$

The source retains q_Y for the K3 and Clifford comparison. Replacing it silently by q_X would invalidate an isometry claim. Local quadratic-form computations and the rational lattice identification are separate from the asserted period rank six. The latter statement was read; its remaining differential proof was not independently audited here.

In the inspected half-sum construction, lines 1794–2371, there are sixteen signed selections above the Gauss map. A coherent assignment satisfies $2h(s) = \sum_{p\in s} p$ and the transition rule $h(s[p]) = h(s) - p$. The assignments form a torsor under the abelian surface's two-torsion. The

source removes local monodromy using fixed selections and holomorphic local halves, then uses the absence of nontrivial etale covers of $\mathbb{P}^1$. An arbitrary independent choice of halves does not provide the needed addition identities.

The two parity classes each have eight selections. The identity $\sum_\sigma \sigma_i\sigma_j = 8\delta_{ij}$ controls the image classes with the degrees of every component included. The addition of the two resulting curve families has total degree eight. Four local preimages over the branch curve each contribute at least two by tangency; equality of the total degree exhausts the possibilities and gives an even *Cartier* pullback of the branch divisor. Separate etale covers of the two curves lift the intervening isogeny and kill the two fundamental-group characters. This produces actual curve domination of the double cover. An abstract rank calculation or an even numerical intersection number would not establish this step.

The selection of generic data is ordered: choose abelian data, then an actual algebraic pencil with good infinity, and only then exclude a countable set of parameters relative to those fixed models and correspondences. The K3-pencil statement through line 2410 was read, but the proofs of integral enlargement, local monodromy, period-stack descent and canonical degree in 2411–3276 remain imported.

7.3 Two substitutions and recovery of the specified tensor

The two-square theorem's contract, lines 3277–3334, was read, as was its application in 4144–4174. Its internal comparison proof in 3335–4143 was not read. It requires the relevant variation, the appropriate projector killing other cohomological degrees, first-substitution uniruled models, and the prescribed order of generic choices. The stated geometric cylinder normalization, including the factor -4, is an imported exact assertion, not something validated by the present finite Clifford calculations.

The inspected recovery argument in 4175–4700 starts from the actual curve correspondence. At a full special-orthogonal construction seed, $W = \mathrm{Cl}^+(U)$ has dimension 128. Over a splitting field the two half-spin modules have dimension eight and each occurs with multiplicity eight. The vector representation appears once in each opposite half-spin tensor product, and not in the same-sign products. Both orientations are needed: the relevant Hom space has dimension $64 + 64 = 128$. The source uses rational Hodge Hom spaces and descent through exact linear kernels; it does not assume that individual complex half-spin summands are rational. Rank-one multiplicity maps span the required component, and divisor correspondences realize the weight-one Hodge maps. Because both factors are odd cohomology, the ordinary tensor flip is minus the graph-induced swap. Our generator criterion is a check of the final specified map, not a replacement for this spanning argument.

The rational K3 transport invokes a full H^2 Hodge isometry, after a quadratic extension by Witt's theorem, and then the algebraicity theorem for that isometry. It does not apply the theorem to an arbitrary finite-rank linear injection.

7.4 All periods and Picard jumps

Lines 4701–5031 keep the rank-eight ambient local system, a prescribed right vector and a compatible polarization fixed even at Picard jumps. Proper relative Hilbert parameter spaces enumerate actual cycles, with rational coefficients and connected components counted. Local Betti constancy is justified for the universal cycle classes. Countably many proper analytic images and Baire's theorem promote the seed construction to an entire period ball. Density of rational orthogonal transformations only places an exact period in a suitable ball; it does not turn nearby tensors into equal tensors.

The final source compression, lines 5032–5255, uses semisimplicity to choose a Hodge retraction,

changes the right vector, and converts the second polarization slot. Theorem 2.1 supplies an explicit canonical choice and proves compatibility along nested jumps. Proposition 3.1 retains the source's distinct second-slot map. It is specifically not a rule to apply the same retraction in both slots. The final algebraic transport still uses abelian homomorphisms up to isogeny, rational K3 isometries and the actual algebraic projector onto T_S.

No mathematical source correction has been established in this scoped study. The form factor two, coherent half sums, degree-eight exhaustion, tensor-flip sign, both half-spin orientations, proper parameter spaces and both polarization slots are retained valid safeguards. The local improvements make some exact finite steps explicit and checkable; they do not certify all imported proofs.

8 Comparison, limitations and further work

8.1 Current primary literature

The public literature refresh was targeted, using public titles and URLs only, on October 8, 2026. Poon's primary abstract and revision metadata [3] describe the Kuga–Satake construction for a Picard-rank-fourteen family and its modular relation with quaternionic abelian eightfolds; the current arXiv version is dated June 15, 2026. The inspected publisher convention in Section 2.3 uses minus cup product, whereas this report follows the assigned source's cup product with a positive period plane. These conventions cannot be interchanged silently. Neither that abstract nor a modular description alone is a certificate for the source's exact algebraic tensor.

The inspected definitions and Corollary 4.7 of Floccari–Fu [4] distinguish an unspecified algebraic embedding from the prescribed Kuga–Satake correspondence. They prove the latter for K3 surfaces associated with projective singular OG6 varieties. That geometric scope does not assert algebraicity for every projective K3 surface. Their inspected discussion reinforces the need for our exact-map criterion, but their full proof was not independently audited.

Buskin's Theorem 1.1 [5], inspected in the primary paper, states algebraicity of rational Hodge isometries between the full second cohomologies of projective K3 surfaces. Its proof is imported. It supplies the stated transport tool, not algebraicity of arbitrary rational maps or of approximate isometries. This limited refresh is not a historical-priority search.

8.2 Private original proof use and attribution

The original confidential manuscript [2] was read in lines 241–307 and 513–694: 249 of 820 lines. Its unchecked complements are 1–240, 308–512 and 695–820. The applied passages concern actual-cycle membership before period inversion, stable finite solves, full-space quotient closure, rational lattice gaps and rational character-orbit projectors. References within those passages do not count as inspected additional originals. No private manuscript title, passage or local path was sent to public search.

The reflection-average candidate for Picard compression was supplied by the project parent and is credited as such. The report independently proves it, its Clifford bimodule property, functoriality and degree-independent denominator bound, and develops the worked lattice and slot certificates. The short-word inverse bound and its full-map test are source-specific local deductions here. Reflection averaging, PBW, module generation and rational-gap principles are classical; no historical novelty is asserted for them.

8.3 Benefits, costs and next steps

The main benefit is reproducibility at the exact-tensor bottleneck: the mask is a coefficient operation, the two slots are explicitly typed, and a finite list of right generators checks the whole Clifford map with an inverse error estimate. Along nested Picard jumps, the masks and second-slot maps compose without a new existential choice. The denominator bound does not grow simply because a higher exterior degree is used.

The main costs are different in kind. Nonadapted frames may be ill conditioned; integral bases may introduce denominators invisible in rational coordinates; the polarization inverse and isogeny transport have additional height costs. None of these improvements constructs the geometric seed cycle, proves the two-square theorem, or replaces the all-period Hilbert argument. The ambient generator optimization uses the specific rational form V_P; the general criterion survives without unit diagonal coefficients, but the signed-word constant need not.

A bounded next step would attach these finite certificates to an explicitly supplied geometric seed correspondence, including its rational cohomology coordinates and integral abelian lattices. Another is to minimize the denominator in an adapted integral Clifford order: (15) shows why an a priori valid bound need not be optimal. A useful computational extension would propagate exact polarization-slot data through a chain of known Picard jumps. None of these tasks should be described as completed until the actual cycles and their arithmetic data are supplied.

9 Independent checks and assessment

The independently executable script `_audit/M062-exact-checks.py` uses Python's exact `Fraction` arithmetic, bit-mask Clifford multiplication, exterior contraction and rational Gaussian elimination. It passed 51,746 assertions in 321 parameter cases:

Finite certificate family	Cases	Assertions
Adapted masks, bimodules and nested composition	16	35,382
Nonadapted masks and denominators	256	7,299
Generator criterion and exact defect bounds	20	8,308
D_4 integral and rational lattice	1	18
Discriminant forms and gluing	16	515
Polarization slots and flag compatibility	12	224
Total	321	51,746

Cases are parameter families; assertions include their exhaustive small PBW subchecks. The tests include a nonmultiplicativity example, a unit-invisible perturbation caught by the generators, and a polarization example where using p twice fails. They are not numerical period experiments, evidence of cycle existence, or a certification of the global Hodge-conjecture conclusion.

The read ledger records the literal intervals and their complements with pre-read and final matching hashes. The structure map records headings separately from reads. The final proof, literature, visual and freeze records belong only to this report. Parent-reported independent checks are not added to the author count.

The contribution rating is C3 for a demonstrated notable local gain: constructive functorial compression with arithmetic control, correctly polarized transport, and a small exact-generator certificate with a quantitative inverse estimate. It is not C4: the construction of algebraic cycles and the source's major geometric comparison remain imported, and classical identities alone are not counted as a new global theorem. This assessment is independent of applicability L3.

References

[1] *Algebraic Kuga–Satake correspondences and Hodge conjectures on a K3 quadratic locus.* Assigned September 30, 2026 public manuscript, pinned local `build/paper.tex`, 5,391 lines. Read scope and captured-source provenance are recorded in the accompanying audit, not inferred from a live branch.

[2] Confidential Ram corpus, `TR-item-011-astra.tex`. Original proof passages 241–307 and 513–694, local consultation only; no public upload.

[3] F. Poon, *Kuga–Satake construction on families of K3 surfaces of Picard rank 14.* arXiv:2504.08270, version 2, June 15, 2026. `https://arxiv.org/abs/2504.08270`. Publisher text: `https://doi.org/10.1002/mana.70173`. Inspected scope: abstract, revision information and the Section 2.3 sign convention.

[4] S. Floccari and L. Fu, *The Hodge conjecture for Weil fourfolds with discriminant 1 via singular OG6-varieties.* Primary author manuscript, Section 2 and Corollary 4.7. `https://irma.math.unistra.fr/~lfu/articles/HCWeilViaOG6.pdf`. Consulted October 8, 2026.

[5] N. Buskin, *Every rational Hodge isometry between two K3 surfaces is algebraic.* arXiv:1510.02852, Theorem 1.1. `https://arxiv.org/abs/1510.02852`. Statement inspected; proof imported.